

OCS-003 — Constitutional Mathematics

Status: Foundational

Authority: Constitutional

Depends on:

- OCS-000 — Constitutional Alphabet
- OCS-001 — Constitutional Grammar
- OCS-002 — Constitutional Logic

Purpose

This specification defines the mathematical structure of Organodynamics.

It does not prescribe numerical mathematics.

It defines constitutional mathematics.

Constitutional Mathematics describes how constitutional meaning may be composed, transformed, preserved, and validated.

Every future specification SHALL remain compatible with this mathematical model.

1. Constitutional Space

Every constitutional construct exists within Constitutional Space.

Constitutional Space is not physical space.

It is the complete set of constitutional entities and constitutional relationships existing under a shared Constitution.

A Constitutional Space SHALL preserve:

Identity

Meaning

History

Relationships

Continuity

2. Constitutional Entity

Every constitutional entity SHALL be represented mathematically as:

$E = (I, M, C, R, H)$

Where:

I = Identity

M = Meaning

C = Context

R = Relationships

H = History

Implementations MAY extend this representation.

Implementations SHALL preserve these components.

3. Constitutional Relationship

A Relationship SHALL be represented as:

$Rel(A, B, T)$

Where

A

Origin Entity

B

Target Entity

T

Relationship Type

Relationship identity SHALL remain independent from participating entities.

4. Constitutional Graph

Every Place SHALL be representable as a constitutional graph.

Nodes represent Constitutional Entities.

Edges represent Constitutional Relationships.

Edges SHALL possess constitutional identity.

Edges SHALL possess constitutional history.

Graph topology MAY change.

Graph meaning SHALL remain reconstructable.

5. Constitutional Distance

Distance SHALL measure semantic separation.

Distance SHALL NOT depend upon implementation.

Distance MAY depend upon:

Shared Context

Shared Evidence

Shared Relationships

Shared History

Equivalent wording SHALL NOT imply minimal distance.

Equivalent meaning MAY.

6. Constitutional Neighborhood

For every constitutional entity,

its Neighborhood SHALL consist of:

Direct Relationships

Indirect Relationships

Inherited Context

Shared Evidence

Related History

Neighborhood SHALL remain computable.

7. Constitutional Connectivity

Connectivity measures constitutional reachability.

Disconnected entities MAY exist.

Disconnected meaning SHALL remain explicit.

Connectivity SHALL NOT imply correctness.

8. Constitutional Path

A Constitutional Path is an explainable sequence of constitutional relationships.

Every path SHALL preserve:

Identity

Traceability

Context

Meaning

Paths SHALL remain replayable.

9. Constitutional Transformation

A Transformation modifies constitutional structure.

Transformation SHALL preserve:

Identity

Meaning

Historical continuity

Traceability

Transformations MAY alter representation.

Transformations SHALL NOT silently alter constitutional semantics.

10. Constitutional Equivalence

Two constitutional structures SHALL be equivalent if:

Meaning is equivalent.

Identity semantics are preserved.

Historical interpretation is preserved.

Relationship semantics are preserved.

Structural equivalence SHALL NOT require identical implementation.

11. Constitutional Composition

Complex constitutional constructs SHALL be compositions of simpler constitutional constructs.

Composition SHALL preserve:

Meaning

History

Identity

Relationships

Composition SHALL remain reversible whenever constitutionally possible.

12. Constitutional Decomposition

Every constitutional construct SHALL be decomposable.

Repeated decomposition SHALL terminate in Constitutional Primitives defined by OCS-000.

No constitutional construct SHALL remain fundamentally opaque.

13. Constitutional Closure

Every constitutional operation SHALL produce a valid constitutional state.

Operations SHALL NOT create undefined constitutional entities.

Closure SHALL preserve constitutional consistency.

14. Constitutional Invariants

The following invariants SHALL hold for every constitutional transformation:

Identity remains stable.

Meaning remains reconstructable.

History remains complete.

Relationships remain valid.

Evidence remains attributable.

Context remains observable.

Violation of any invariant SHALL invalidate the transformation.

15. Constitutional Metrics

Implementations MAY calculate constitutional metrics.

Metrics SHALL remain informational.

Metrics SHALL NOT redefine constitutional meaning.

Examples include:

Relationship Density

Perspective Coverage

Evidence Coverage

Semantic Stability

Knowledge Connectivity

Interpretation Diversity

Metrics SHALL remain explainable.

16. Constitutional Optimization

Optimization SHALL preserve constitutional meaning.

Optimization MAY improve:

Performance

Storage

Communication

Latency

Builder Experience

Optimization SHALL NEVER reduce constitutional integrity.

17. Constitutional Completeness

A constitutional mathematical model SHALL be considered complete when:

Every constitutional entity is representable.

Every relationship is representable.

Every transformation is representable.

Every explanation is reconstructable.

Every history is replayable.

Every constitutional specification derives consistently from this model.

Conformance

An implementation conforms to OCS-003 if independent computational models preserve equivalent constitutional mathematics despite differing internal

architectures.

Constitutional Principle

Mathematics exists to preserve constitutional meaning.

Not to replace it.

Every equation defined within Organodynamics is meaningful only insofar as it preserves the ability to reconstruct understanding across time, implementations, and participants.

End of Specification OCS-003